MADFU · MERCHANT ONBOARDING

Unified Onboarding Master Flow Board

Every screen, decision, branch, retry and outcome of the implemented onboarding — left to right. Screens are the live prototype components; journeys are derived by running the real journey engine. Synthetic data only.

24 screen nodes
18 distinct screens
20 decisions
101 connections
29 journeys

LEGEND

- ☐ User-facing screen (live wireframe)
- ▣ External integration (paid / simulated)
- ⬚ Future scope
- - - - Retry loop
- ◇ Business decision
- ⬚ Manual review / flag
- ⬭ Connector (jump)
- ······ Permitted step back
- ▨ Backend operation
- ☐ Final outcome
- ── Transition

How to read: start at **S01** (new applicant) or **S00** (returning). The flow splits after the category decision into Path A (Registered Business) and Path B (Freelancer). Applications leaving self-onboarding drop into the manual-review lane; every route ends in an outcome.

ASSUMPTIONS & OPEN DECISIONS — not confirmed business rules

D-01 SAR 100,000 boundary — Configurable in rules/config.ts (REVENUE_BANDS, REVENUE_ROUTING).
D-04 OTP resend & expiry — Expiry 120 s, cooldown 30 s, 3 resends, resend does not reset attempts; lockout requires a new application.
D-07 Existing freelancer — add certificate — Documented branch from S01-X to S01-F placeholder marked FUTURE FLOW. No implementation.
D-10 IBAN technical errors — 3 technical errors → status "pending (provider)", separate flag; contract allowed subject to processing rules.
D-13 "Pending manual review" category — Stops and saves a review lead (Outcome C) before any paid call. Configurable: STOP_AS_LEAD | CONTINUE_WITH_FLAG.
D-16 Certificate not found after Nafath — Flag CERT_NOT_FOUND_MANUAL_ENTRY (blocking).
D-19 Nafath presentation — Dedicated screen with a very large circular number.
D-22 IBAN ownership provider — Typed adapter returns verified / not_verified / technical_error.

D-02 Contract signing technology — Generic checkbox + "Accept and submit" placeholder; acceptance simulated and not legally binding.
D-05 Category restriction catalogue — Clearly labelled fixture catalogue in rules/categories.ts.
D-08 API failures after retries — 2 silent auto-retries, then 3 manual retries, then "save and exit"; resuming grants a fresh retry allowance.
D-11 Nafath technical failure — NAFATH_POLICY.technicalFailureConsumesAttempt = false (configurable).
D-14 Identity lock — Fields read-only; a new application is required. Abuse via restarting needs a server-side control.
D-17 Lead resubmission — Demo registry in browser storage (clearable from Prototype Controls).
D-20 Nafath status delivery — Waiting state persists across refresh; resolution simulated by timer or dev simulator.
D-23 VAT number rule — 15 digits starting and ending with 3 (structural placeholder).

D-03 Government validation patterns — Structural regex placeholders (10 digits) kept separate from simulated government verification.
D-06 Certificate / category reconciliation — Fixture mapping table; mismatches flagged (journey continues); expired / inconsistent certificates flagged and contract disabled (manual review).
D-09 Contract decline handling — Application saved, not submitted, status CONTRACT_PENDING with CONTRACT_DECLINED flag; applicant can review again.
D-12 IBAN check for CR without Nafath — Not called (cost avoidance; application goes to manual review). Configurable per applicant type.
D-15 Resume authentication — Browser-storage resume without re-authentication (demo only).
D-18 Category picker — Real catalogue size may require a searchable / grouped picker instead of two dropdowns.
D-21 Related-party data — Separate mock call; entity type field drives the individuals-only filter.

Shared entry & eligibility
Both applicant types · entry → duplicate checks → OTP → revenue → category
START · new applicant (both types)
↻ Validation errors
S01 Start application
Both
Next · fields valid
DECISION
Identity already registered?
Check 1 · CR / National ID
No
BACKEND
Check 2 · mobile linkage
informational · never blocks
Mobile not yet verified
INTEGRATION
Send SMS OTP
simulated
Same mobile already verified in this application → skip OTP
↻ Incorrect (< 3) · expired → resend
S02 Mobile OTP
Both
Verify
DECISION
Code correct?
max 3 incorrect
Verified
3rd incorrect → locked
DECISION
Revenue routing
CR range 100K–20M
Self-onboarding
↻ Selection missing
S03 Business category
Both
Next
DECISION
Category eligibility
before paid checks
↩ Start a new application
START · returning applicant
S00 Resume application
Both
Continue application
↪ Continue at the last saved step
Yes · stop (mobile check skipped)
S01-X Existing account
Both
Freelancer · add certificate
FUTURE FLOW · pending specification
S01-F Add another certificate
Freelancer
S02-L Verification locked
Both
Registered Business · allowed / CR-only
CR below SAR 100K or above SAR 20M
Restricted · pending review
Freelancer · allowed
Freelancer · CR-only subcategory
Path A — Registered Business (CR)
CR verification → authorized person → Nafath → details → contract
INTEGRATION
Government CR lookup
paid call
↻ Unavailable / error → retry (auto ×2, manual ×3)
S04 Business verification
Business
DECISION
CR result
Valid
Expired
↩ Not found
↩ Edit CR number → S01
MANUAL REVIEW
Flag CR_EXPIRED
continue · review later
Continue
BACKEND
Related parties → individuals only
↻ Data unavailable → try again
S05 Authorized person
Business
Continue
DECISION
Eligible individual selected?
None on CR
MANUAL REVIEW
No eligible individual
entities only on CR
One selected
↻ Technical failure → retry (no attempt used)
S06 Nafath (authorized person)
Business
Send request
INTEGRATION
Nafath request
paid call · 2-digit number
Approve in Nafath
DECISION
Nafath result
Verified
Rejected / timed out
DECISION
Attempts used (max 2)
shared counter
↩ 1 of 2 used · retry same person
↩ 1 of 2 used · choose another person (counter kept)
2 of 2 used
MANUAL REVIEW
Nafath unverified
manual verification · no contract
Continue (no contract)
Manual identity verification
↻ 1st failure / error → correct or retry
S08 Business details
Business
Next · fields valid
DECISION
Call IBAN ownership API?
Nafath verified?
Yes
INTEGRATION
IBAN ownership API
match: registered business
DECISION
Ownership result
Verified
Failed ×2 · outage ×3
No · collect without API
MANUAL REVIEW
IBAN unverified / pending
flagged · does not block contract
Continue
DECISION
Contract eligible?
Nafath is decisive
Eligible
Not eligible → submit for manual review
S09 Merchant agreement
Business
DECISION
Accept or decline?
Accept
Decline → confirm
S09-D Agreement not accepted
Business
↩ Review agreement
BACKEND
Record acceptance
simulated
Path B — Freelancer
Nafath on National ID → certificates → details → contract
MANUAL REVIEW
Flag CR-only category
continue · business decides
Continue
↻ Technical failure → retry (no attempt used)
S06 Nafath (National ID)
Freelancer
Send request
INTEGRATION
Nafath request
paid call · 2-digit number
Approve in Nafath
DECISION
Nafath result
Verified
Rejected / timed out
DECISION
Attempts used (max 2)
↩ 1 of 2 used · retry
2 of 2 used
INTEGRATION
Certificate lookup
paid call
↻ Service unavailable → try again
S07 Certificate selection
Freelancer
Continue
DECISION
Certificate available?
None found / all attached
S07-M Manual certificate entry
Freelancer
↻ Invalid / already linked
Continue
MANUAL REVIEW
Certificate entered manually
manual review · no contract
Continue (no contract)
One selected
BACKEND
Reconcile with Madfu category
DECISION
Reconciliation result
Match
Mismatch / expired / inconsistent
MANUAL REVIEW
Certificate flagged
mismatch / expired / inconsistent
Continue
Restricted mapping
↻ 1st failure / error → correct or retry
S08 Business details
Freelancer
Next · fields valid
DECISION
Call IBAN ownership API?
Nafath verified?
Yes
INTEGRATION
IBAN ownership API
match: National ID
DECISION
Ownership result
Verified
Failed ×2 · outage ×3
No · collect without API
MANUAL REVIEW
IBAN unverified / pending
flagged
Continue
DECISION
Contract eligible?
Nafath is decisive
Eligible
Not eligible → submit for manual review
S09 Merchant agreement
Freelancer
DECISION
Accept or decline?
Accept
Decline → confirm
S09-D Agreement not accepted
Freelancer
↩ Review agreement
BACKEND
Record acceptance
simulated
Submitted
Manual review & sales leads
Applications leaving self-onboarding
MANUAL REVIEW
Revenue outside CR range
lead · no paid calls
MANUAL REVIEW
Restricted / review category
lead · no paid calls
MANUAL REVIEW
Certificate maps to restricted
lead · cost recorded
BACKEND
Save sales / review lead
SALES_LEAD
BACKEND
Submit for manual verification
Path B · APPLICATION_SUBMITTED
BACKEND
Submit for manual verification
Path A · APPLICATION_SUBMITTED
Lead confirmation
Received
Received
Final outcomes
What the applicant sees (internal statuses stay internal)
END
O-C Interest received (sales lead)
Both
Thank you for your interest in Madfu.
END
O-A Submitted · Freelancer
Freelancer
Your application has been submitted successfully.
END
O-B Received · Freelancer
Freelancer
Thank you. Your application has been received.
END
O-A Submitted · Registered Business
Business
Your application has been submitted successfully.
END
O-B Received · Registered Business
Business
Thank you. Your application has been received.